David Guo

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EDUCATION

University of Toronto **Sep 2026 – 2028 (Expected)**
M.A.Sc. in Information Engineering **Supervisor:** Prof. Scott Sanner

University of Toronto **Sep 2021 – Apr 2026**
B.A.Sc. in Engineering Science (Machine Intelligence) **GPA:** 3.85/4.0, with Honours
Coursework: Deep Learning, Probabilistic Models, Optimization, Computer Vision, Reinforcement Learning.

PUBLICATIONS & PREPRINTS

*Equal contribution

ICLR Submission Liang, J.*, Gallagher, L.*, Chen, K.*, **Guo, D.***, Toroghi, A., Liu, Y. S., & Sanner, S.
Have I Scene This Before? Spatially Grounded Conversational Memory for Complex Queries in Egocentric Assistants. Submitted to ICLR.

UMAP '26 **Guo, D.***, Sun, M.*, Jiang, Y.*, Liang, J., & Sanner, S.
VOGUE: A Multimodal Dataset for Conversational Recommendation in Fashion.

ACL '26 Findings Liang, J.*, Liu, Y.*, **Guo, D.***, Sun, M., Jiang, Y., & Sanner, S.
Evaluating Scene-based In-Situ Item Labeling for Immersive Conversational Recommendation.

RESEARCH EXPERIENCE

Data-Driven Decision Making (D3M) Lab, University of Toronto **May 2025 – Present**
Researcher | Supervisor: Prof. Scott Sanner

- Co-designed and ran a human–human conversational recommendation study, producing VOGUE: 60 dialogues and 900 turns with annotations of recommendation behavior and information needs.
- Developed stage-aware analyses and LLM-based evaluation methods for conversational recommendation; led empirical analysis for VOGUE, resulting in co-first-authored papers at ACM UMAP and Findings of ACL 2026.
- Wrote an undergraduate thesis developing an information-needs perspective on conversational recommendation, synthesizing findings from the VOGUE study and broader analysis of user–assistant interaction.
- Co-developed Ego-SpaCR, a conversational spatial-memory benchmark with 620 video sessions, 95 conversations, and 3,100 queries requiring joint reasoning over dialogue and 3D object relationships; Our method (SpaCMEM) achieved the highest evaluated answer accuracy, a 22% relative gain over the strongest baseline.
- Developing a world-state–centric AR/XR research framework with persistent object state, query/history interfaces, subscriptions, and synchronized RGB-D/pose replay for reproducible perception and agent research.

Matter Lab, University of Toronto **Sep 2025 – Sep 2026**
Research Student | PI: Prof. Alan Aspuru-Guzik

- Developed a permutation-invariant graph neural network for solubility prediction across single- and multi-solvent systems, targeting generalization to unseen solutes and solvent compositions.
- Built the end-to-end modeling stack across approximately 200,000 experimental measurements, including molecular graph construction, mixture representations, uncertainty-aware training, and OOD evaluation protocols.
- Designed ablations and distribution-shift experiments to isolate the contributions of molecular descriptors, message passing, and mixture representation; led analysis and manuscript development.

Acceleration Consortium, University of Toronto **Apr 2026 – Sep 2026**
Machine Learning Research Student | Supervisor: Dr. Nasim Abdollahi

- Led ML development for a self-driving nanoemulsion laboratory, building GNN prediction/evaluation pipelines and model interfaces for iterative experiment selection.
- Integrated the predictive model with Bayesian optimization for closed-loop autonomous experiment selection, enabling initial end-to-end self-driving laboratory runs.

RESEARCH PROJECTS

Normative Spinal Geometry Modeling for Metastatic Disease **Fall 2025**
Capstone Research Project | *Sunnybrook Research Institute* | *PI: Dr. Michael Hardisty*

- Combined Generalized Procrustes Analysis and statistical shape modeling with gradient-boosted regression to reconstruct patient-specific normative spinal geometry, achieving under 6 mm global 3D RMSE.
- Integrated the resulting models into a Python library and visualization dashboard for assessing metastasis-related changes in spinal geometry.

INDUSTRY EXPERIENCE

ModiFace (L'Oréal) | *Software Developer Intern – Augmented Reality* **May 2024 – Apr 2025**

- Re-architected and shipped components of the Vue-based integration layer for an AR makeup platform, eliminating redundant camera and image-augmentation paths across legacy libraries.
- Reworked error and exception handling end to end across the integration layer, standardizing failure paths, telemetry, and diagnostics for production AR applications.

DealerVu | *Software Developer* **May 2023 – Aug 2023**

- Shipped end-to-end internationalization for the SuperSOM dealership platform, implementing dynamic language loading, locale-aware formatting, adaptive React layouts, and supporting .NET/SQL infrastructure.

AWARDS & HONOURS

NSERC Undergraduate Student Research Award (USRA) Summer 2025
Dean's List, University of Toronto 2022–2026
Rowe Innovation Award, Praxis design project — $4,000 2022
University of Toronto Scholars Entrance Award — $7,500 2021

RESEARCH & TECHNICAL SKILLS

Conversational AI & HCI: Conversational recommendation, human–AI interaction, user studies.
Agents & intelligent systems: Agent orchestration, retrieval-augmented systems, persistent memory, world state.
Research & evaluation: Experimental design, benchmarks, ablations, OOD evaluation, error analysis.
Machine learning: GNNs, language models, Bayesian optimization, representation learning, segmentation.
Frameworks & tools: PyTorch, scikit-learn, RDKit, OpenCV, NumPy, SciPy, Git, Docker, Linux.
Languages: Python, C/C++, C#, MATLAB, JavaScript/TypeScript, SQL.